

Personalized Federated Learning with Gaussian Processes: A Reproduction-Oriented Analysis with PathMNIST Extension

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Project: <https://github.com/kennyliyang/pFedGP-PathMNIST>

Abstract

Personalized federated learning (PFL) aims to exploit cross-client collaboration without collapsing heterogeneous clients into a single global decision rule. This paper presents a reproduction-oriented analysis of Personalized Federated Learning with Gaussian Processes (pFedGP), focusing on how its shared deep-kernel representation and client-specific Gaussian-process classifiers support personalization, uncertainty-aware prediction, and scalability through GP-Tree and inducing-point variants. The reproduced run shows validation accuracy improving from 0.3992 to a best observed value of 0.6247, with final test accuracy 0.6261 and test loss 1.1384. We interpret this run as evidence of coherent optimization rather than as a full multi-seed benchmark reproduction. In addition, we incorporate a supplementary PathMNIST experiment as domain-transfer evidence: it moves the empirical discussion from natural-image benchmarks toward medical histopathology classification while keeping calibration, robustness, and out-of-distribution claims explicitly bounded by the available evidence.

1. Introduction

Federated learning trains models over distributed clients while limiting data movement. A standard global model is attractive when client distributions are broadly aligned, but it is fragile when clients differ in class coverage, sample size, noise, acquisition process, or domain. Personalized federated learning addresses this failure mode by allowing local adaptation while still exploiting federation-scale statistical strength.

pFedGP [10] is a representative PFL method built around a useful separation of responsibilities. A shared neural feature map learns a deep kernel representation across clients, while each client maintains a Gaussian-process classifier in

that learned embedding space. This design is conceptually well matched to non-IID settings: the federation learns a transferable similarity metric, whereas the GP classifier preserves client-specific decision behavior and probabilistic uncertainty.

This paper is not a new algorithmic contribution. Its goal is to convert available reproduction artifacts, original paper evidence, and a new PathMNIST extension into a disciplined empirical argument. We distinguish direct evidence from reproduced runs, paper-reported benchmark context, and supplementary implementation evidence. This distinction is important because a single optimization trace cannot support the same claims as multi-seed benchmark tables, calibration diagrams, or held-out-client evaluations.

Our contributions are threefold. First, we analyze the reproduced pFedGP run as an optimization-behavior check, emphasizing the relation between validation loss, validation accuracy, best-checkpoint progression, and held-out test performance. Second, we relate those findings to the original paper’s claims about representation sharing, client-local GP inference, calibration, and scalability. Third, we add PathMNIST as a supplementary medical-image experiment, positioning it as external-validity evidence rather than as proof of clinical robustness.

2. Methodological Background

2.1. Personalized Federated Learning

PFL methods vary in how they divide global and local responsibility. Some methods share a representation and train local heads; others regularize local models toward a global solution, use meta-learning, cluster clients, or generate personalized parameters. Classical FedAvg [1] learns a single global model, whereas representative personalized methods such as FedPer [2], pFedMe [3], and pFedHN [4] introduce client-specific adaptation. pFedGP belongs to the representation-sharing family, but replaces a deterministic classifier head with client-local Gaussian-process inference.

2.2. Deep-Kernel Gaussian Processes

A Gaussian process defines a distribution over functions through a kernel [5]. In low-data local regimes, this is useful because predictions naturally encode uncertainty. A client-only GP, however, is limited by the quality of its kernel and by the amount of local data available to fit it. pFedGP mitigates this by learning a neural feature map $g_\theta(\cdot)$ across the federation and applying a kernel in the learned embedding space, following the broader deep-kernel-learning idea of combining neural feature extractors with nonparametric kernels [6]. Thus, clients share representation learning but retain local Bayesian decision rules.

Let a client have local data $\mathcal{D}_i = \{(x_j, y_j)\}_{j=1}^{n_i}$. After the shared encoder maps inputs to embeddings $z_j = g_\theta(x_j)$, a GP prior is placed on the latent function,

$$f_i(z) \sim \mathcal{GP}(0, k_\phi(z, z')), \quad (1)$$

where k_ϕ is a kernel in the learned representation space. For a test embedding z_* , define $K_i = [k_\phi(z_a, z_b)]_{a,b=1}^{n_i}$ and $k_{i,*} = [k_\phi(z_1, z_*), \dots, k_\phi(z_{n_i}, z_*)]^\top$. Under a Gaussian observation model, the joint distribution follows the standard GP form also used in the Bayesian active-learning reference equations (3)–(5) [7]:

$$\begin{bmatrix} \mathbf{y}_i \\ f_i(z_*) \end{bmatrix} \sim \mathcal{N}\left(0, \begin{bmatrix} K_i + \sigma^2 I & k_{i,*} \\ k_{i,*}^\top & k_\phi(z_*, z_*) \end{bmatrix}\right). \quad (2)$$

Conditioning on the local observations gives the predictive posterior

$$p(f_i(z_*) | \mathcal{D}_i, z_*) = \mathcal{N}(\mu_i(z_*), \sigma_i^2(z_*)), \quad (3)$$

with

$$\mu_i(z_*) = k_{i,*}^\top (K_i + \sigma^2 I)^{-1} \mathbf{y}_i, \quad (4)$$

$$\sigma_i^2(z_*) = k_\phi(z_*, z_*) - k_{i,*}^\top (K_i + \sigma^2 I)^{-1} k_{i,*}. \quad (5)$$

Equations (3)–(5) make explicit why GP personalization is attractive, and they are also the reason GP models are widely used in uncertainty-driven active learning [8, 9]: the posterior mean acts as a client-specific prediction rule, while the posterior variance provides an uncertainty estimate tied to the client’s local evidence. In pFedGP, these equations are applied within a classification-oriented GP-Tree approximation rather than as a standalone scalar regression head.

2.3. GP-Tree and Inducing Points

For multiclass classification, pFedGP uses a GP-Tree decomposition rather than a single monolithic multiclass GP. Each tree node is a binary GP classifier, and a multiclass prediction is obtained by following a path through the tree. The method also includes inducing-point variants, such as

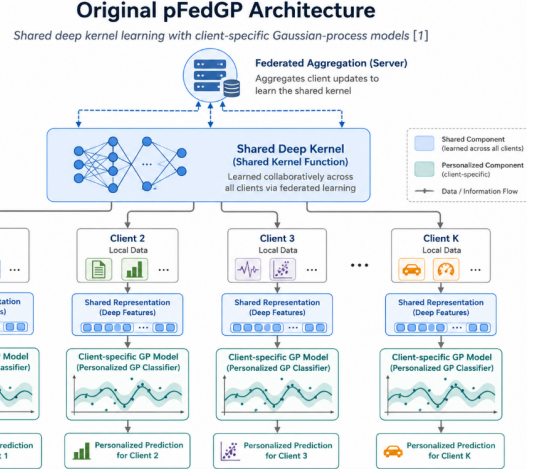


Figure 1. Original pFedGP architecture. The shared neural feature map learns a kernel representation across clients, while each client retains a local GP classifier.

pFedGP-IP-data and pFedGP-IP-compute, to reduce inference cost and improve efficiency when local sample sizes or compute budgets are constrained. More generally, GP scalability is often handled through sparse or approximate kernel machinery; random Fourier features are a representative approximation for large-scale kernel methods [11].

Figure 1 summarizes the architectural claim: personalization is not achieved by merely fine-tuning a global classifier. Instead, global representation learning and local Bayesian inference are separated.

3. Reproduced Optimization Evidence

3.1. Evidence Policy

The reproduced run provides direct numerical evidence for validation loss, validation accuracy, best-validation progression, test loss, and test accuracy. Paper-reported benchmark tables provide broader comparison context, while the PathMNIST repository provides supplementary implementation and domain-transfer evidence. We do not mix these evidence levels as if they came from the same experimental protocol.

3.2. Learning Behavior

The validation loss decreases rapidly at the beginning and then enters a slower stabilization phase. This pattern is consistent with an optimization process that first captures transferable structure and then performs lower-yield refinement. The corresponding validation accuracy increases from 0.3992 to a best observed value of 0.6247. The best-so-far curve is smoother than raw validation accuracy because it tracks the model-selection envelope rather than round-level fluctuations.

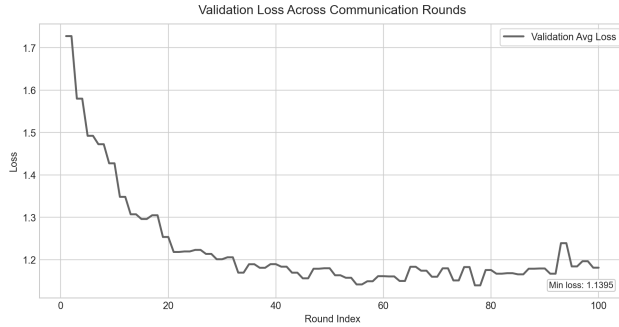


Figure 2. Validation loss trajectory across communication rounds.

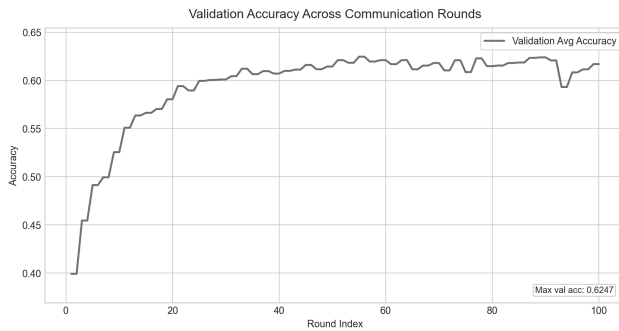


Figure 3. Validation accuracy trajectory. Accuracy improves substantially from the initial state and then fluctuates around a higher-performance regime.

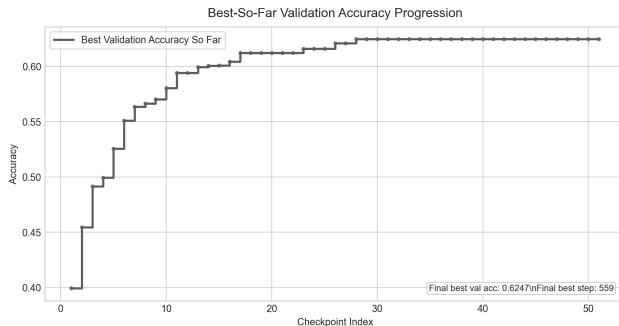


Figure 4. Best-so-far validation accuracy. The envelope shows cumulative model-selection improvement despite non-monotonic round-level behavior.

The final test accuracy is close to the best validation accuracy. This supports the limited claim that the selected model transfers reasonably to the held-out evaluation in this run. It does not, by itself, establish multi-seed stability, baseline superiority, calibration, robustness, or OOD generalization.

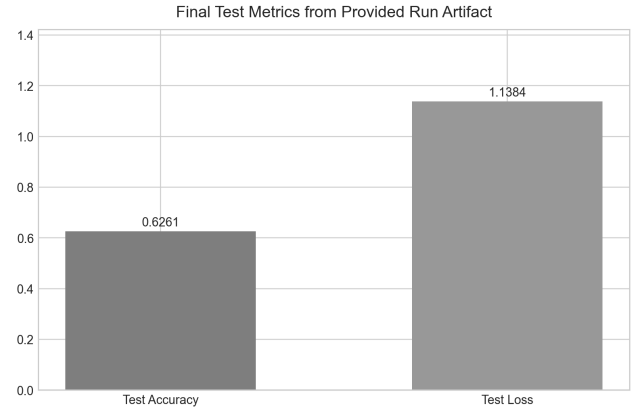


Figure 5. Final test metrics from the reproduced run. Test accuracy is 0.6261 and test loss is 1.1384.

Table 1. Direct reproduced-run metrics.

Metric	Value
Initial validation accuracy	0.3992
Best validation accuracy	0.6247
Best step	559
Test accuracy	0.6261
Test loss	1.1384

Table 2. Benchmark evidence reported by [10].

Dataset	FedAvg	FedPer	pFedMe	pFedHN	pFedGP
CIFAR-10 (50)	56.4	83.8	86.4	90.2	89.9
CIFAR-10 (100)	59.7	81.5	85.0	87.4	88.8
CIFAR-100 (50)	23.6	48.3	49.8	60.0	61.2
CIFAR-100 (100)	24.0	43.6	47.7	52.3	59.8

3.3. Paper-Reported Benchmark Context

Table 2 summarizes benchmark evidence reported by the original paper. The important pattern is not that pFedGP wins every cell, but that personalized methods outperform FedAvg under heterogeneous data and that pFedGP is especially competitive in harder CIFAR-100 settings.

4. Supplementary PathMNIST Experiment

The PathMNIST experiment is best interpreted as a supplementary domain-transfer experiment, not as a replacement for the original benchmark suite. Its value is that it moves the empirical discussion from generic natural-image datasets toward medical histopathology image classification, which is closer to the privacy-sensitive medical motivation for PFL.

The linked implementation adapts pFedGP to the PathMNIST subset of MedMNIST [12, 13]. It describes a pipeline for generating a 100-client dictionary-format dataset, run-

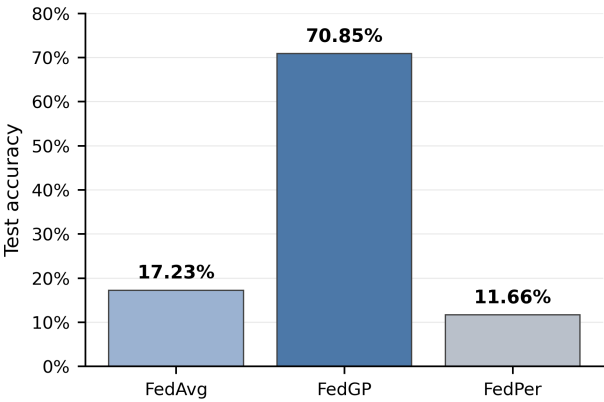


Figure 6. Final test accuracy comparison on the PathMNIST supplementary experiment. FedGP achieves the best observed accuracy (70.85%), outperforming FedAvg (17.23%) and FedPer (11.66%). Results are taken from the project run logs; source implementation: <https://github.com/kennyliyang/pFedGP-PathMNIST>.

ning pFedGP variants, and comparing against FedAvg and FedPer baselines. This supports the report’s application-level claim: the pFedGP architecture can be evaluated in a medical-image PFL workflow without changing its basic decomposition into shared deep-kernel learning and client-local GP inference.

Figure 6 summarizes the final test accuracies obtained from the supplementary PathMNIST runs in the linked project. In the corrected comparison, FedGP attains the strongest result, reaching 70.85% test accuracy, whereas FedAvg obtains 17.23% and FedPer obtains 11.66%. This result is useful because it separates the domain-transfer claim from the performance claim: the repository demonstrates an executable medical-image PFL workflow, while the observed accuracy comparison indicates that the current FedGP configuration is substantially stronger on the PathMNIST run.

This evidence must remain carefully bounded. If the experiment lacks matched multi-seed baseline tables, per-client metrics, calibration scores, and held-out-client or corruption tests, it cannot justify claims about clinical robustness or universal medical generalization. Its strongest defensible role is external-validity evidence: pFedGP remains applicable to a medical pathology classification task and can be evaluated in the same PFL framework as the original method.

Table 3. Role of the supplementary PathMNIST experiment.

Role	Contribution
Dataset diversity	Adds histopathology image classification beyond natural-image benchmarks.
Application relevance	Connects the medical-FL motivation to an executable MedMNIST pipeline.
Personalization stress test	Can test heterogeneous client behavior if the 100-client split is non-IID.
Evidence limitation	Does not prove calibration, OOD robustness, or clinical reliability without dedicated metrics.

5. Discussion

5.1. Calibration and Robustness

The original pFedGP paper emphasizes predictive calibration. Reliability diagrams, expected calibration error, maximum calibration error, and Brier score are the proper evidence types for this claim. The reproduced run used here does not include confidence-bin data or predicted probability distributions, so calibration cannot be reproduced numerically from the supplied artifact.

The same limitation applies to noisy-input robustness and out-of-distribution client generalization. These claims require corruption protocols, novel-client splits, baseline outputs, and per-client evaluation records. The GP component provides a plausible mechanism for improved uncertainty behavior because posterior uncertainty is part of the predictive model. However, mechanism plausibility is weaker than direct empirical reproduction.

5.2. Claim-Evidence-Reasoning View

The central claim of pFedGP is not merely that a model obtains a higher accuracy number. The deeper claim is that the architecture resolves a tension in PFL: clients need models adapted to local distributions, but individual clients often have too little data to learn strong representations alone. The shared deep kernel addresses representation learning, while the local GP addresses personalization and uncertainty.

The reproduced optimization curves support the claim that the implementation can learn a useful model under the tested setting. The paper-reported benchmark evidence supports the broader comparative claim under heterogeneous image-classification tasks. The PathMNIST extension supports external validity by introducing a medical-image domain. None of these evidence items alone proves all claims;

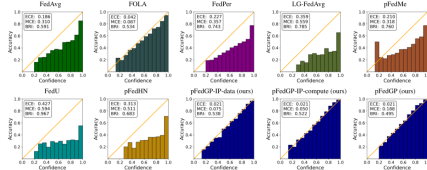


Figure 3: Reliability diagrams on CIFAR-100 with 50 clients. Diagonal indicates perfect calibration. Each plot also shows the expected & maximum calibration error (ECE & MCE) and the Brier Score (BRI). Lower is better.

the training data take an active part in the GP inference formulas (Eq. 6), while in pFedGP-IP-data the data impact in a weak manner only through the loss function. (5) pFedGP-IP-data is especially helpful when few samples per class are available, e.g., CIFAR-100 with 500 clients. That last point is further illustrated by decoupling the effect of the number of clients from that of the training set size. To illustrate that, in Appendix E.2 we fixed the number of clients and varied the number of training samples per class. From this experiment, we deduced that both factors (individually) contribute to

Figure 7. Reliability-diagram evidence from the original paper, included as paper-reported calibration context.

together, they form a bounded empirical argument.

6. Conclusion

pFedGP is a methodologically compelling PFL approach because it combines federation-scale representation learning with client-local Bayesian classifiers. The reproduced run supports a coherent learning trajectory and reaches 0.6261 test accuracy, while the original paper provides benchmark, calibration, and OOD context. The supplementary PathMNIST experiment strengthens the external-validity argument by extending the empirical setting to medical histopathology classification. The correct interpretation is therefore qualified: the available artifacts support learning behavior and application relevance, while stronger claims about calibration, robustness, and clinical reliability require additional dedicated evidence.

7. Team Contributions

This report is submitted by **GROUP1**. Table 4 summarizes the project division of labor. This section is included for the course/project-report version; for a strictly anonymous CVPR review submission, it should be removed or anonymized.

Table 4. Team contribution summary.

Member	Role and responsibility
Liyang Luo	<i>Model Architect.</i> Implemented the core GP algorithm, the Pólya-Gamma augmentation module, and the mathematical implementation of the inducing-points FITC optimization.
Mingxuan LI	<i>Data Engineer.</i> Handled MedMNIST data preprocessing, implemented the non-IID data-partitioning logic, and constructed the DKL feature-extraction CNN.
Feiyang Huang	<i>Federated Infrastructure.</i> Developed the server-client communication logic, implemented weight-aggregation strategies, and set up the FedAvg baseline model.
Luo li	<i>Evaluation & Metrics.</i> Tracked experimental results, computed uncertainty metrics including ECE, MCE, and Brier score, generated reliability diagrams, and formatted the final report.

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